AP PRECALCULUS

Unit 1 • Worksheet 1.9-1.10B

Worksheet 1.9-1.10B • Rational Functions and Asymptotes

Name: _____ Date: _____

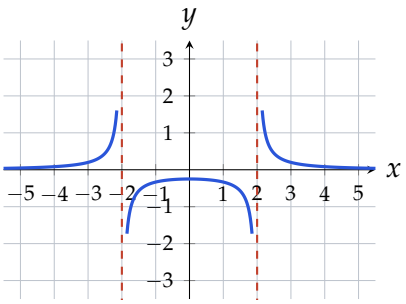
Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The graph of $r(x) = \dfrac{1}{x^2 - 4}$ is shown. Which statement is true?



(A) $\lim_{x \to 2^-} r(x) = \infty$

(B) $\lim_{x \to 2^+} r(x) = -\infty$

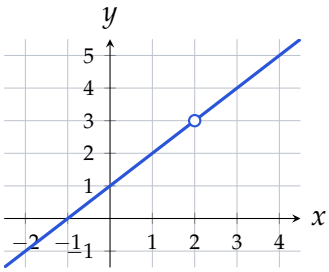
(C) $\lim_{x \to -2^+} r(x) = \infty$

(D) $\lim_{x \to 2^-} r(x) = -\infty$

Question 2

NO CALCULATOR

The graph of a rational function is a line with a hole at $(2,3)$. Which could be its equation?

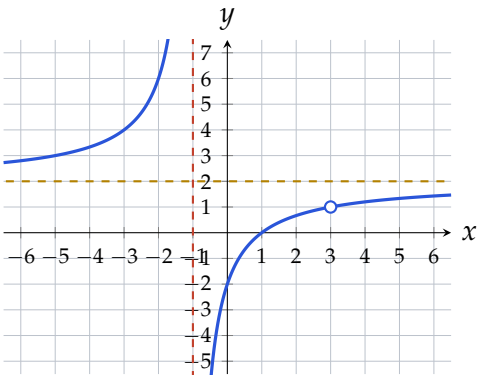


(A) $\dfrac{x^2-4}{x-2}$
(B) $\dfrac{x+1}{x-2}$
(C) $\dfrac{x^2-x-2}{x-2}$
(D) $x+1$

Question 3 (Free response)

NO CALCULATOR

The graph of r is shown. It has a vertical asymptote $x=-1$, a horizontal asymptote $y=2$, a zero at $x=1$ and a hole at $(3,1)$.



(a) Write limit statements describing r near $x=-1$, at $x=3$, and as $x \to \pm\infty$.
(b) Write an expression for $r(x)$ that has all of these features.
(c) Solve $r(x) < 0$.